

Teacher Guide — middle / module-14- magic-squares

IKS Curriculum

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Overview

Module 14 — Magic Squares (*Bhadragaṇita*)

Age band: Middle · **Grades:** 6–8 · **Duration:** 10 sessions × 45 min

Learning objectives

By the end of this module, students will be able to:

1. **Define** a magic square as an $n \times n$ grid of distinct integers whose rows, columns, and main diagonals share a common sum (the *magic constant*).
2. **Derive** the magic constant formula $S = n(n^2 + 1)/2$ for an $n \times n$ square filled with the integers 1 to n^2 .
3. **Construct** any odd-order magic square (3×3 , 5×5 , 7×7) using Nārāyaṇa Paṇḍita's stair-step (*turagagati*) method.
4. **Verify** that a given grid is or is not magic by direct row, column, and diagonal sum-checks.
5. **Recognise** the historical 4×4 *Khajuraho* magic square and explain why its panel-magic property goes beyond the basic definition.
6. **Locate** *Bhadragaṇita* (Book 14 of Nārāyaṇa Paṇḍita's *Gaṇita-kaumudī*, 1356 CE) in the chronology of Indian mathematics, alongside the Chinese *Lo Shu* and Dürer's 1514 European square.

NEP 2020 alignment

- **§4.6** — Integration of Indian Knowledge Systems: students engage Nārāyaṇa Paṇḍita's algorithmic method as a piece of mainstream mathematical content, not a curiosity.
- **§4.23** — Experiential learning: discovery-by-trial of the 3×3 square, paired construction of larger squares, and a design-your-own project.
- **§11.8** — Holistic, multidisciplinary education: cross-cultural comparison (Lo Shu, Khajuraho, Dürer) and connections to combinatorics, Sudoku, and recreational mathematics.

- §4.27 — Local context: the Khajuraho inscription is a real Indian artefact (Lakṣmaṇa temple complex, ~10th–11th c. CE) that students can locate geographically and visually.

Prerequisites

- Confident arithmetic with single- and double-digit addition.
- Comfortable identifying rows, columns, and diagonals in a grid.
- Familiarity with the integers 1 to 25 and small multiplication facts.

Key vocabulary

See `_shared/glossary.md` (Module 14 section).

- *Bhadragaṇita* — “mathematics of auspicious squares”; Book 14 of the *Gaṇita-kaumudī*
- *Gaṇita-kaumudī* — Nārāyaṇa Paṇḍita’s 1356 CE mathematical compendium
- *samānaṅka* — equal-sum property (the magic constant)
- *turagagati* — “horse-step,” Nārāyaṇa’s stair-step construction method
- *yantra* — diagram (used here only to acknowledge the cultural context, not to teach ritual)

Materials checklist

- Printed blank 3×3, 4×4, and 5×5 grids (~5 per student across the module)
- Coloured pencils or markers (3 colours per student)
- A class set of numbered tiles 1–25 (or paper squares) — at least one per pair
- Printed image of the *Khajuraho* magic square panel for Day 6
- Printed image of Dürer’s *Melencolia I* (1514) — only the magic-square corner needed
- Stopwatch / wall clock for Day 5 timed construction

Day-by-day overview

Day	Title	Focus	Assessment
1	What is a magic square?	Discover the 3×3 by trial	Self-check
2	The magic constant	Derive $S = n(n^2 + 1)/2$ for $n = 3$	Formative quiz Part A
3	How many 3×3 squares are there?	Symmetry and counting	Pair worksheet
4	The stair-step method	Nārāyaṇa’s <i>turagagati</i> on the 3×3	Activity rubric
5	A 5×5 magic square	Apply <i>turagagati</i> to $n = 5$	Formative quiz Part B
6	The 4×4 and Khajuraho	Even-order squares, panel-magic property	Discussion

Day	Title	Focus	Assessment
7	Who was Nārāyaṇa?	<i>Gaṇita-kaumudī</i> in context	Reflection
8	Cross-cultural — Lo Shu and Dürer	Comparison, project planning	Project draft
9	Sudoku connection + project work	Modern recreational connection	Project draft
10	Summative + share	Quiz and design-your-own showcase	Summative + project

Citation sources used

- **Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī* (Book 14, *Bhadragaṇita*), 1356 CE** — the primary historical source for the construction methods.
- **Kim Plofker, *Mathematics in India* (Princeton University Press, 2009)** — secondary scholarly overview; chapter on later medieval mathematics.
- **T. Hayashi (various papers on history of Indian mathematics)** — cited as supporting context in `sources.md`.

Full chain in `sources.md`.

Project brief (one-line teaser)

Each pair designs an original magic square (5×5 or larger) using the stair-step method, presents it as a labelled poster or short demo, and writes one paragraph explaining the construction. Full brief in `assessments/project-brief.md`.

10-Day Lesson Plan

Module 14 (Middle) — 10-Day Lesson Plan

At a glance

Day	Session title	Lesson file	Activity / assessment
1	What is a magic square?	<code>days/day-01-what-is-a-magic-square.md</code>	Discovery worksheet
2	The magic constant	<code>days/day-02-magic-constant.md</code>	Formative quiz Part A
3	How many 3×3 squares are there?	<code>days/day-03-counting-and-</code>	Pair worksheet

Day	Session title	Lesson file	Activity / assessment
		symmetry.md	
4	The stair-step method	days/day-04-stair-step-method.md	activities/activity-01-stair-step-drill.md
5	A 5×5 magic square	days/day-05-five-by-five.md	Formative quiz Part B
6	The 4×4 and Khajuraho	days/day-06-khajuraho.md	Class discussion
7	Nārāyaṇa and <i>Gaṇita-kaumudī</i>	days/day-07-narayana.md	Reflection paragraph
8	Lo Shu and Dürer	days/day-08-cross-cultural.md	Project planning + activities/activity-02-cross-cultural-compare.md
9	Sudoku + project session	days/day-09-sudoku-and-project.md	Project draft
10	Summative + share	days/day-10-summative-and-share.md	quizzes/summative.md + project

Pacing notes

- Days 1–2 are the conceptual core. Don’t rush the *discovery* of the 3×3 — most of the lasting motivation comes from a student building one themselves.
- Day 4 introduces the *turagagati* method on the 3×3 (which students already know). This lets the algorithm be the focus, not the construction.
- Day 5 applies the same algorithm to a 5×5. If the class struggled with Day 4, push Day 5 by one day and combine it with Day 6.
- Day 6 (4×4 Khajuraho) is the cultural-context highlight. It is NOT required that students build a 4×4 themselves at Middle band — that’s a Senior task. Showing the inscription and counting its panel-magic property is enough.
- Days 8–10 are project + presentation. Light enough to combine if school calendar requires.

Daily warm-up (recurring)

Every class starts with **2 minutes of “row sum recall”**: teacher writes any 3 numbers from a known magic square; students name the 4th number that completes the row to 15. (E.g. “1, 5, ?” → 9.) Builds arithmetic fluency.

What students leave the module with

- One personally constructed 5×5 magic square they can show a friend.
- The formula $S = n(n^2 + 1)/2$ memorised and explained.

- An understanding that magic squares appeared independently in multiple cultures — and that Nārāyaṇa Paṇḍita (1356 CE) produced the most systematic pre-modern treatment.
- A poster, demo, or short write-up as their project artefact.

Teacher Notes

Teacher Notes — Module 14 Magic Squares (Middle)

Posture

This module is about **pattern + procedure**. The pattern is the magic constant (a row, column, or diagonal always sums to the same number); the procedure is Nārāyaṇa’s stair-step algorithm for constructing any odd-order square. Both are accessible to Middle-band students. The cultural anchor is the 1356 CE *Gaṇita-kaumudī*; the math is the math.

Honesty about history: magic squares were known in many cultures (China, India, Islamic world, Europe). Nārāyaṇa Paṇḍita’s *Bhadragaṇita* is the most systematic pre-modern treatment we have in writing, but he was not the sole or first discoverer. Frame him as a brilliant systematiser, not as a lone genius.

Common misconceptions

1. **“The magic constant is 15 for every magic square.”** Only for the 3×3 with numbers 1–9. For an $n \times n$ magic square with $1..n^2$, the constant is $S = n(n^2 + 1)/2$. So $5 \times 5 \rightarrow 65$, $4 \times 4 \rightarrow 34$, $7 \times 7 \rightarrow 175$.
2. **“There are many different 3×3 magic squares with 1–9.”** There is *one* such square up to symmetry — 8 if you count rotations and reflections. This is a beautiful surprise for students on Day 3.
3. **“The stair-step method only works on odd squares.”** Correct — and that’s a feature, not a bug. The 4×4 needs a different method (which we show but do not teach in detail at Middle band).
4. **“The Khajuraho square is just decorative.”** It is unusually strong: it is panel-magic (every 2×2 block also sums to 34), not just row/column/diagonal magic. This is a real mathematical property; the panel sums are an extra structural constraint beyond the basic definition.
5. **“Magic squares are mystical/protective and must be treated reverently.”** Some traditional uses are protective/contemplative; in this curriculum we engage them as combinatorial structures. We acknowledge the cultural context without using ritual framing.

Differentiation hints

- **Tier up:** Construct a 7×7 magic square (same method, just more cells). Verify the constant equals 175. Bonus: try a “rotated start” — does the algorithm still produce a magic square if you begin in a different cell?

- **Tier down:** Stay with the 3×3 . Practise the formula on smaller mental targets. Use coloured tiles for the stair-step rather than writing.

Differential pacing

- A class strong in arithmetic and pattern-recognition can compress Days 1–3 into 2 days and use the extra time for the 7×7 challenge.
- A class new to grids should NOT skip Day 3 (counting and symmetry). Symmetry intuition feeds the rest of the module.

Sensitivity notes

- Some students may have heard magic squares described as *yantras* in religious or astrological contexts at home. Acknowledge: “yes, they have been used that way historically; in this class we study them as mathematics.” Don’t dismiss the home context; do keep the classroom frame mathematical.
- Avoid the framing “ancient Indians knew this first.” Multiple cultures encountered magic squares; Nārāyaṇa’s contribution is the *systematic* treatment, not the discovery.

Project oversight

- Three project formats (poster, demo video, written exposition) — see [assessments/project-brief.md](#). Each requires the student to actually *construct* a magic square at least 5×5 .
- Common project pitfall: students copy a 5×5 from the internet without using the *turagagati* method. Insist on showing the construction order step-by-step.
- Verification step: every project must include the row/column/diagonal sum-check table. No exceptions — this is what makes it a *mathematical* project rather than a craft project.

Citation discipline

This module cites a Sanskrit primary source (Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14, 1356 CE) and a modern scholarly secondary source (Plofker 2009). Students should cite the primary source in their project work. Middle-band students give a paraphrase plus citation, not a Sanskrit quote.

Visualising the stair-step rule

The *turagagati* algorithm for an odd-order $n \times n$ square:

1. Place 1 in the **top-middle** cell.
2. From the cell containing k , move **one up and one to the right** (diagonally up-right) to place $k + 1$.
3. If you go off the top, wrap to the bottom row.
4. If you go off the right, wrap to the left column.
5. If the destination cell is **already filled**, place $k + 1$ in the cell **directly below** the cell containing k instead.
6. Repeat until all n^2 cells are filled.

The most common error students make is at step 5: when blocked, they go to the wrong “fallback” cell. Drill this until they can recite “blocked → directly below.”

If you only have 6 days

- Day 1 (combined: what is it + magic constant)
- Day 2 (counting and symmetry — quick)
- Day 3 (stair-step on 3×3)
- Day 4 (stair-step on 5×5)
- Day 5 (Khajuraho + Nārāyaṇa context)
- Day 6 (summative + brief share — no full project)

Sources

Sources — Module 14 Magic Squares (Middle)

Primary source

Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī* (1356 CE) — Book 14: *Bhadragaṇita* (“Mathematics of Auspicious Squares”).

- The *Gaṇita-kaumudī* is a Sanskrit mathematical compendium in 14 books (*vyavahāras*). Book 14, *Bhadragaṇita*, is dedicated to magic squares and is the most extensive pre-modern Indian treatment of the topic.
- Nārāyaṇa presents construction methods for: (a) odd-order squares via a stair-step / *turagagati* placement, (b) even-order squares of orders divisible by 4 via a complement-pair method, and (c) “singly even” orders (6, 10, 14...) via a modified method.
- For Middle band we use only the odd-order method, since it is fully accessible with a single rule.

Cite as: Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE.

Secondary scholarly source

Kim Plofker, *Mathematics in India* (Princeton University Press, 2009).

- Chapter 8 (“The School of Mādhava”) and surrounding context cover 14th-century Indian mathematics. Magic squares are discussed in the context of Nārāyaṇa’s combinatorial work.
- Used as the bridge between Sanskrit primary source and a modern teacher-friendly summary.

Cite as: Plofker (2009), *Mathematics in India*.

Additional historical context (teacher reference)

- **Hayashi, T.** Multiple papers in *Historia Mathematica* and *Indian Journal of History of Science* on the chronology and content of pre-modern Indian mathematics, including Nārāyaṇa's work on combinatorics and magic squares.
- **Sarma, S. R.** Work on Indian astronomical and mathematical instruments; sometimes discusses *yantra* diagrams that include magic squares.
- These are background reading for the teacher; not cited to students directly.

Material artefacts

- **Khajuraho 4×4 magic square inscription.** Found at the Khajuraho temple complex (Madhya Pradesh, India); dated *circa* 10th–11th c. CE — the exact date is debated. The square is known as a “panel-magic” square (also called “pan-diagonal” or “most-perfect”): rows, columns, broken diagonals, *and* every 2×2 panel sum to the same constant.
- **Lo Shu (China).** Earliest secure attestation in the *Da Dai Liji* (Han dynasty, ~5th–1st c. BCE). Legend attributes the discovery to Emperor Yu (~2000 BCE) seeing a turtle whose shell carried the pattern — that legend is *not* taken as historical evidence here.
- **Albrecht Dürer, *Melencolia I* (1514).** The print's upper-right corner contains a 4×4 magic square where the bottom-middle two cells encode the date “1514.”

What to tell students about the citation chain

The construction methods we teach are real and ancient: Nārāyaṇa's *turagagati* algorithm is exactly the same algorithm modern textbooks now call the *Siamese method* or the *de la Loubère method* — but Nārāyaṇa described it nearly three centuries before de la Loubère reported it from his 1687 embassy to Siam. We will cite Nārāyaṇa as the historical first systematic exposition we have in writing. We will *not* claim he was the only inventor, nor the first ever — there are earlier hints in Indian, Chinese, and Arabic sources. The honest framing is: “Nārāyaṇa gave us the most complete pre-modern systematic treatment we have.”

Day 1 — What is a Magic Square?

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) state the definition of a magic square, and (b) construct a 3×3 magic square with the numbers 1 to 9 by their own trial.

Materials

- Printed blank 3×3 grid (3 per student, with the digits 1–9 listed beside the first grid)
- A loose set of paper tiles or small cards numbered 1–9 (one set per pair)
- Whiteboard

Vocabulary introduced today

- *magic square* — a square grid in which every row, every column, and both main diagonals add up to the same total.
- *magic constant* — that common total (sometimes also called the *magic sum*).

Lesson flow

Warm-up (5 min)

- Teacher draws an empty 3×3 grid on the board. “*I want to place the numbers 1, 2, 3, 4, 5, 6, 7, 8, 9 in this grid — each number exactly once — so that **every row, every column, and both diagonals add up to the same number**. Can it even be done?*”
- Take 2–3 quick guesses. Don’t confirm. Write on the side of the board: “Possible? If yes — what’s the magic sum?”

Core (20 min)

1. **Hand out tiles 1–9 and the blank grids.** Pairs work for 8 minutes to try to build one.
2. **Mid-pair check-in (after 4 min):** “*What number is in the centre of your best attempt so far?*” Most pairs will have 5 in the centre. Ask why.
3. **Give a hint, not the answer.** “*The middle number is 5. Each row sums to 15. Now try again.*”
4. **Reveal one valid square** on the board (after most pairs have struggled — 7–8 min in):

2		7		6

9		5		1

4		3		8

Verify together: each row $\rightarrow 15$. Each column $\rightarrow 15$. Both diagonals $\rightarrow 15$. “*That sum 15 is called the magic constant of this square.*”

5. **Definition on the board.** Students copy this into their notebooks:

A **magic square** is an $n \times n$ grid filled with distinct numbers such that every row, every column, and both main diagonals add up to the same total — the **magic constant**.

Activity (15 min) — Verify and vary

- **Part A (5 min):** Each student picks any 3 lines (rows, columns, or diagonals) of the revealed square and verifies the sum on their notebook.
- **Part B (10 min):** Pairs try to make a *different* 3×3 magic square using 1–9. (Without saying so, you are previewing Day 3 — they will discover symmetries.) Most will produce rotations or reflections of the same square. That is the discovery, not the failure.

Wrap-up (5 min)

- Quick show of hands: “Did anyone find a square that is genuinely different from the one on the board?” (Most will say yes; some of those “different” squares will be rotations.)
- Preview Day 2: “*Tomorrow we’ll prove why the magic constant is exactly 15 for the 3×3 with 1–9. We won’t guess — we’ll derive it.*”

Student-facing content

A **magic square** is an $n \times n$ grid of distinct numbers where every row, every column, and both main diagonals add up to the same total — the **magic constant**.

The simplest non-trivial magic square uses the numbers 1 to 9 in a 3×3 grid. Its magic constant is **15**.

2		7		6

9		5		1

4		3		8

Verify: rows $\rightarrow 2+7+6=15, 9+5+1=15, 4+3+8=15$. Columns $\rightarrow 2+9+4=15, 7+5+3=15, 6+1+8=15$. Diagonals $\rightarrow 2+5+8=15, 6+5+4=15$.

Homework

- On graph paper, copy the magic square shown in class. Then, beside it, draw three different “rotations” of it (turn the page $90^\circ, 180^\circ, 270^\circ$). Are they still magic squares? Verify one row of each.

Differentiation

- **One tier up:** Try to make a 4×4 magic square using the numbers 1 to 16. (Hard! The magic constant is 34. You will likely fail without a method. That’s fine — we’ll learn the method later.)
- **One tier down:** Practise just verifying the revealed 3×3 square. Sum every row, column, and diagonal slowly and carefully.

Teacher notes

- Resist the urge to give the answer in the first 5 minutes. The discovery is the lesson.
- Some students will ask “is it always 15?” Answer: “for THIS square, with THESE numbers (1–9). Tomorrow we’ll see why.”
- A few students will try to use the same number twice. Reinforce that all 9 numbers must be distinct — that’s part of the definition.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — the historical source for the systematic treatment of magic squares we will follow this module. Today we set the stage; Day 4 begins his explicit method.

Day 2 — The Magic Constant

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) derive the magic constant $S = n(n^2 + 1)/2$ for the 3×3 case by direct argument, and (b) apply the formula to predict the constant for $n = 4, 5, 6, 7$.

Materials

- Whiteboard
- Notebook + pen
- The 3×3 magic square from Day 1, visible on the board

Vocabulary review

- *magic square, magic constant* — defined Day 1.

Lesson flow

Warm-up (5 min)

- Recall chant: “ 3×3 square with 1 to 9 — magic constant is ____?” Class: “15.”
- “Why exactly 15, though? Why not 14 or 17?” Take 2 student responses.

Core (20 min)

1. **The sum argument.** Write on the board:

Add up all the numbers in the square: $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 = 45$.

“This total goes into the square once. Now — the square has 3 rows. Each row has the same sum S . So $3 \times S = 45 \rightarrow S = 15$.”

Repeat the logic with student input. They should arrive at the formula step-by-step.

2. **Generalise.** For an $n \times n$ square containing the numbers 1, 2, ..., n^2 :

Sum of all numbers = $1 + 2 + \dots + n^2 = n^2(n^2 + 1) / 2$

The square has n rows, each summing to S . So:

$$n \times S = n^2(n^2 + 1) / 2 \rightarrow \mathbf{S = n(n^2 + 1) / 2}$$

3. **Apply.** Compute together for several n :

n	n^2	n^2+1	$n(n^2+1)/2$
3	9	10	$3 \times 10 / 2 = \mathbf{15}$
4	16	17	$4 \times 17 / 2 = \mathbf{34}$
5	25	26	$5 \times 26 / 2 = \mathbf{65}$
6	36	37	$6 \times 37 / 2 = \mathbf{111}$

n	n^2	n^2+1	$n(n^2+1)/2$
7	49	50	$7 \times 50 / 2 = 175$

4. **A paraphrase from the source.** Nārāyaṇa Paṇḍita, in *Bhadragaṇita* (Book 14 of *Gaṇita-kaumudī*, 1356 CE), gives the magic-constant rule explicitly: the common sum is one n -th of the sum of the numbers in the square. We’ve just re-derived his rule. (See *sources.md*.)

Activity (15 min) — Formative quiz Part A

- Hand out quizzes/formative.md Part A.
- Students complete individually. 10 minutes; 3 minutes for self-check against the board answer key.
- Part A is 6 short questions: 3 on the formula, 3 on verifying given squares.

Wrap-up (5 min)

- “Quick check: what would the magic constant be for a 10×10 magic square with the numbers 1 to 100?” Class works it out: $S = 10 \times 101 / 2 = 505$.
- Preview Day 3: “Tomorrow we ask: how many different 3×3 magic squares are there with 1 to 9? Take guesses tonight.”

Student-facing content

The magic constant formula.

For an $n \times n$ magic square containing the numbers 1, 2, ..., n^2 :

$$S = n(n^2 + 1) / 2$$

Why? The sum of all n^2 numbers is $n^2(n^2 + 1) / 2$ (the standard sum-to- n formula). This total is split equally among the n rows, each of which sums to S . So $n \times S = n^2(n^2 + 1) / 2 \rightarrow$ divide both sides by n .

n	Constant
3	15
4	34
5	65
6	111
7	175

Nārāyaṇa Paṇḍita stated this rule in *Bhadragaṇita* (1356 CE).

Homework

- Verify the formula for $n = 8$ and $n = 9$. (Answers: 260 and 369.)
- For each, also compute the total sum $n^2(n^2 + 1)/2$ directly and confirm it equals $n \times S$.

Differentiation

- **One tier up:** What if the magic square contains *consecutive* numbers but starting at k , not at 1? (E.g. a 3×3 with 5, 6, 7, ..., 13.) Derive a formula for the new magic constant. (Answer: $S = n \times k + n(n^2-1)/2$.)
- **One tier down:** Focus on the 3×3 case alone. Verify the constant 15 by summing every row, column, and diagonal of the Day-1 square. Do it slowly.

Teacher notes

- A common stumble: students compute $1 + 2 + \dots + 9$ incorrectly. Remind them that $1+9$, $2+8$, $3+7$, $4+6$ each = 10, plus 5 = 45. This itself is a pretty argument worth pausing on.
- For the generalisation, some students will struggle with the n^2 notation. Walk them through the $n = 5$ case explicitly: $1 + 2 + \dots + 25 = 25 \times 26 / 2 = 325$; split among 5 rows $\rightarrow 65$.
- The formula $S = n(n^2 + 1)/2$ is **the** rule of this module. Drill it.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — paraphrased here: the magic constant equals one n -th of the total sum of the n^2 entries.

Day 3 — How Many 3×3 Magic Squares Are There?

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) discover that the 3×3 magic square with 1–9 is **unique up to symmetry**, (b) list the 8 symmetries (4 rotations + 4 reflections), and (c) explain why 5 must occupy the centre.

Materials

- Printed 3×3 grids (5 per pair) with the digits 1–9 listed
- Coloured pencils (3 colours per student)
- The Day-1 square visible on the board

Vocabulary introduced today

- *symmetry* — a way of moving or flipping a figure so that it looks the same in a different orientation.
- *rotation* — turning around a centre by 90° , 180° , or 270° .
- *reflection* — flipping across a line (horizontal, vertical, or one of the diagonals).

Lesson flow

Warm-up (5 min)

- Daily fluency: “row sums to 15 — give me the missing number.” Quick 5 rounds.
- Recall: “Yesterday we proved the magic constant for a 3×3 is 15. Today’s question: how many different 3×3 magic squares with 1–9 are there?”
- Take 3 student guesses. Write the guesses on the board.

Core (20 min)

1. **Why 5 must be in the centre.** Argue together:

- The centre cell sits on 4 lines: the middle row, the middle column, and both diagonals.
- The four corner cells each sit on 3 lines (a row, a column, and one diagonal).
- The four edge-middle cells each sit on 2 lines (a row and a column).
- Sum of all lines = $4 \times 15 = 60$ (4 lines through the centre? — actually 8 lines total in a 3×3 : 3 rows + 3 columns + 2 diagonals. Their sum is $8 \times 15 = 120$.)
- In that sum of 120, each cell is counted as many times as the number of lines it sits on: centre counted 4 times, corners 3 times each, edges 2 times each.
- Total: $4c + 3(\text{corner sum}) + 2(\text{edge sum}) = 120$, where c is the centre value.
- Corner sum + edge sum + $c = 45$ (the total of 1..9). So $3(\text{corner+edge}) + 4c + \dots$ — work it out together. The cleanest answer: $c = 5$.

(Don’t make them complete the full algebra — show the structure, conclude $5 = \text{centre}$.)

2. **Why 2, 4, 6, 8 must be at corners.** The four corner cells must contain the four *even* numbers from 1–9 (which are 2, 4, 6, 8). Reason: corners sit on diagonals — and the diagonals also sum to 15. With 5 in the centre, each diagonal pair must sum to 10. Pairs from 1–9 (excluding 5) summing to 10 are (1,9), (2,8), (3,7), (4,6). The two diagonals use four of these — pick any two pairs. The remaining four numbers (the odd or even ones, depending on the pair choice) end up at the edges. Most cleanly: even numbers at corners.

3. **The 8 symmetries.** Demonstrate on the board. Take the Day-1 square and:

- Rotate 90° , 180° , 270° — three new arrangements.
- Reflect horizontally, vertically, and across each diagonal — four more.
- Total: 8 different-looking arrangements, all the *same* magic square up to symmetry.

Theorem (state, don’t prove): There is exactly **ONE** 3×3 magic square with 1–9, *up to symmetry*. Counting all 8 rotated/reflected versions, you get 8 grids that all share the same essential structure.

Activity (15 min) — Find all 8

- Each pair starts with the Day-1 square. They must produce all 8 rotated/reflected versions on their own paper.

- Verify each by checking one row sums to 15.
- Pair share: any two pairs swap, and compare — did they get the same 8?

Wrap-up (5 min)

- “There is essentially *ONE* 3×3 magic square with 1–9. Eight different orientations of the same square. Tomorrow we ask: is there a method to construct magic squares of any size — not just 3×3 ?”
- Preview Day 4: Nārāyaṇa’s stair-step method.

Student-facing content

There is only ONE 3×3 magic square with 1–9 — up to symmetry.

The 4 rotations and 4 reflections together produce 8 grid arrangements. They all share the same structure: - 5 is in the centre. - The four even numbers (2, 4, 6, 8) are at the four corners. - The four odd numbers other than 5 (1, 3, 7, 9) are at the four edge-middles.

2 7 6	6 7 2	← reflection (horizontal flip)
-----	-----	
9 5 1	1 5 9	
-----	-----	
4 3 8	8 3 4	

Eight arrangements; one square.

Homework

- Take the 3×3 magic square and ask: if I change every entry by adding 10 (so the square becomes 12, 17, 16, 19, 15, 11, 14, 13, 18), is it still a magic square? What’s the new magic constant? Verify one row.

Differentiation

- **One tier up:** Try the same question for 4×4 magic squares — how many are there? Look it up (or estimate). The answer is 880 distinct 4×4 squares up to symmetry; that’s a surprise.
- **One tier down:** Just memorise: 5 in the centre, even at corners. Use coloured pencils to highlight the pattern in the Day-1 square.

Teacher notes

- The full proof that 5 must be central is a beautiful counting argument but takes 10 minutes if done thoroughly. Choose: either do it well, or do it briefly with hand-waves. Don’t do it half-fast — that confuses more than it clarifies.
- The “eight symmetries form a group” idea is the *dihedral group* D_4 . Don’t introduce that name at Middle band — note it for yourself; it returns at Senior band.
- A few students will produce a “9th” square that turns out to be one of the 8 in disguise. Walk them through which symmetry transforms theirs into the original.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — discusses equivalence classes of magic squares (paraphrased).

Day 4 — The Stair-Step Method (*turagagati*)

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will reproduce Nārāyaṇa Paṇḍita’s stair-step (*turagagati*) algorithm to construct a 3×3 magic square from scratch, and will be able to state all five steps of the rule.

Materials

- Whiteboard
- Printed blank 3×3 grids (3 per student)
- A printed copy of the five-step rule (handout) — see Student-facing content below

Vocabulary introduced today

- *turagagati* (IAST: turaga-gati; lit. “horse-step” or “knight-move”) — Nārāyaṇa Paṇḍita’s name for the diagonally-upward-right placement rule for constructing odd-order magic squares.

Lesson flow

Warm-up (5 min)

- Daily fluency: row-sum-to-15 quick fire. 5 rounds.
- “You’ve all built the 3×3 by guessing. Today we replace guessing with a method — Nārāyaṇa Paṇḍita’s *turagagati* rule from 1356 CE.”

Core (20 min)

1. **State the rule.** Five steps. Write them on the board. Have students copy.

The stair-step (*turagagati*) method — for any odd-order $n \times n$ square:

1. Place **1** in the **top-middle** cell.
2. From the cell containing k , move **one up and one to the right** (diagonally up-right) to place $k + 1$.
3. If you go off the **top edge**, wrap to the **bottom row** (same column you’d have landed in).
4. If you go off the **right edge**, wrap to the **left column** (same row).
5. If the destination cell is **already filled**, place $k + 1$ in the cell **directly below** the cell containing k instead.

Continue until all n^2 cells are filled.

2. **Demonstrate live on the 3×3.** Walk through cell-by-cell, narrating each step:

Step:	1	2	3	4	5
Cell:	(1, c2)	(off-top,	(off-right,	filled –	(off-
	top,				top,
		wrap to	wrap to	go directly	wrap
	to				
		bottom)	left col)	below)	
	bottom)				

The result:

. 1 .	. 1 .	. 1 .	. 1 .
2 1 .			
-----	-----	-----	
. . .	. . .	3 . .	3 . .
→ 3 . .	→		
-----	-----	-----	
. . .	. . 2	. . 2	4 . 2
4 . 2			

Continue: 6 (up-right from 5, wraps), 7 (up-right from 6, blocked → below), 8, 9. Final:

2 7 6

9 5 1

4 3 8

“Notice — this is exactly the square we discovered on Day 1. The method gave it to us directly, no guessing.”

3. **Why does this work?** Brief preview: the method spreads numbers evenly across rows, columns, and diagonals in a structured way. The full *proof* is Senior-band material. For now, the algorithm produces a magic square; verify by summing.

Activity (15 min) — Drill

- Use `activities/activity-01-stair-step-drill.md`.
- Each student constructs three 3×3 squares from scratch — first slowly, then faster.
- Pair check: did your neighbour get the same square? Verify magic constant = 15.

Wrap-up (5 min)

- Quick recap of the 5-step rule.
- Preview Day 5: *“Tomorrow we apply the same algorithm to a 5×5. The rule does not change.”*

Student-facing content

The stair-step (*turagagati*) method — for any odd-order $n \times n$ magic square:

1. Place **1** in the **top-middle** cell.
2. From the cell with k , move **one up and one to the right** to place $k + 1$.
3. If you go **off the top**, wrap to the **bottom row**.
4. If you go **off the right**, wrap to the **left column**.
5. If the destination is **already filled**, place $k + 1$ **directly below** the cell containing k instead.

2		7		6

9		5		1

4		3		8

Nārāyaṇa Paṇḍita described this method in *Bhadragaṇita* (1356 CE). Modern textbooks sometimes call it the *Siamese method* or *de la Loubère method* — named for a European traveller who reported it from Siam in 1687. Nārāyaṇa described it 300+ years earlier.

Homework

- Construct three 5×5 magic squares using the stair-step method. The magic constant should be 65 for all three. Yes — they will all be the same square. We'll discuss why tomorrow.

Differentiation

- **One tier up:** Try the method starting at the *bottom-middle* with the number 1 going *down-right* instead of up-right. Does it still produce a magic square? (Yes — a rotated version. Symmetry strikes again.)
- **One tier down:** Re-do the 3×3 construction with the rule sheet open in front of you. Speed comes later.

Teacher notes

- The most common student error is step 5 — when blocked, students go in random directions instead of “directly below.” Drill this. Repeat: “blocked → directly below.”
- A second common error: students try to apply this method to a 4×4 square. Remind them this is **odd-order only**. The 4×4 needs a different method (Day 6).
- When students wrap (step 3 or 4), they often “lose” themselves on the grid. A useful trick: imagine the grid tiled — the cell at (off-top, column c) is the same as (bottom row, column c).
- Walk around during the activity and correct misplacements early; mistakes compound.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — primary source for the *turagagati* method we are using.

- Plofker (2009), *Mathematics in India* — secondary scholarly context.

Day 5 — A 5×5 Magic Square

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will independently construct a 5×5 magic square using the *turagati* method and verify its magic constant of 65.

Materials

- Printed blank 5×5 grids (3 per student)
- The 5-step rule handout from Day 4
- Stopwatch (for timed second-construction)

Vocabulary review

- *turagati* — stair-step method (introduced Day 4).
- *magic constant* for 5×5 → **65** (from Day 2).

Lesson flow

Warm-up (5 min)

- Class recall: 5-step rule, recited together.
- Magic-constant check: “For a 5×5 with numbers 1–25, the magic constant is ____?” (65.)
- Pre-class question: did everyone produce the *same* 5×5 last night? Most will say yes.

Core (20 min)

1. **Build a 5×5 together on the board.** Narrate every step. Take 8–10 minutes — go cell-by-cell.

Start with 1 in row-1, column-3 (the top middle). Place 2 up-right (off the top → wraps to bottom row, column 4). Place 3 up-right from 2 (off the right → wraps to row 4, column 5)... continue.

Expected result (the classical Nārāyaṇa 5×5):

17		24		1		8		15

23		5		7		14		16

4		6		13		20		22

10		12		19		21		3

 11 | 18 | 25 | 2 | 9

2. **Verify together.** Pick:

- One row $\rightarrow 17 + 24 + 1 + 8 + 15 = 65 \checkmark$
- One column $\rightarrow 17 + 23 + 4 + 10 + 11 = 65 \checkmark$
- The main diagonal $\rightarrow 17 + 5 + 13 + 21 + 9 = 65 \checkmark$
- The anti-diagonal $\rightarrow 15 + 14 + 13 + 12 + 11 = 65 \checkmark$

3. **Notice the structure.** Take 3 minutes to observe:

- 13 (the middle number) is in the centre — consistent with the $n^2/2 + 1 = 13$ rule for odd-order.
- Numbers 1–9 occupy a “broken diagonal” pattern; 16–25 occupy the complementary pattern.
- The same algorithm that gave us the 3×3 gave us this. The rule does not change with size.

Activity (15 min) — Formative quiz Part B + speed construction

- **First 10 min:** Hand out quizzes/formative.md Part B. Students complete individually.
- **Last 5 min:** Each student starts a fresh 5×5 grid. On “Go,” they construct a magic square from scratch as fast as they can. Self-time. Verify one row before stopping the watch.

Wrap-up (5 min)

- Show of hands: who finished in under 4 minutes? Under 6? Under 8?
- “Yesterday’s home practice was the same 5×5. Tomorrow we look at a different beast — the 4×4.”
- Preview Day 6: Khajuraho and the 4×4.

Student-facing content

The 5×5 magic square — classical Nārāyaṇa form

Magic constant: 65 (from $S = n(n^2 + 1)/2$ with $n = 5$).

17 | 24 | 1 | 8 | 15

 23 | 5 | 7 | 14 | 16

 4 | 6 | 13 | 20 | 22

 10 | 12 | 19 | 21 | 3

 11 | 18 | 25 | 2 | 9

Built using the *turagagati* method, exactly as for the 3×3 — the rule does not change with size, as long as n is odd.

Homework

- Construct a 7×7 magic square using the stair-step method. The magic constant should be 175. Yes, it takes time — work patiently. Bring it tomorrow for spot-checks.

Differentiation

- **One tier up:** Construct a 9×9 magic square. Constant = 369. Time it.
- **One tier down:** Construct another 5×5 , slowly, with the rule sheet visible. Speed is not the goal yet.

Teacher notes

- For the live demonstration, use a slow pace and check student-following frequently — “where would 2 go?” “is that cell filled?”
- Students may resist because “I already built one at home.” Push them to *re-build* it on the board — narrating the rule reinforces it.
- A few students will produce mirror-image squares because they reversed the up-right direction. Show that mirror images are also valid magic squares (by symmetry from Day 3) — but consistency matters when you’re describing the algorithm.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadraṇita*), 1356 CE — the 5×5 example is one of the worked examples in the original text.

Day 6 — The 4×4 Square and Khajuraho

Module: Magic Squares (*Bhadraṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) recognise the Khajuraho 4×4 magic square, (b) verify its rows, columns, diagonals, and 2×2 panels all sum to 34, and (c) explain why this “panel-magic” property is *stronger* than the basic magic-square property.

Materials

- Printed image of the Khajuraho inscription panel (or a clean diagram of the square)
- Printed copy of the Khajuraho 4×4 grid (one per student) for sum-checking
- Coloured pencils

Vocabulary introduced today

- *Khajuraho* — site in Madhya Pradesh, India, with a temple complex dating to roughly the 10th–11th c. CE (Chandela dynasty). One temple wall carries a numbered 4×4 grid inscription.

- *panel-magic square* (also called *pandiagonal* or *most-perfect* in modern terminology) — a magic square where, in addition to rows, columns, and main diagonals, **every 2×2 panel** sums to the magic constant, and so do the **broken diagonals**.

Lesson flow

Warm-up (5 min)

- “Yesterday you built a 5×5 from scratch. The same algorithm doesn’t work for 4×4 — odd-order only. Today we look at one famous 4×4 — from a temple wall in central India, roughly 10th–11th c. CE.”
- Show the Khajuraho image. Don’t analyse yet. Just let students look.

Core (20 min)

1. **The Khajuraho 4×4.** Project or hand out:

7		12		1		14

2		13		8		11

16		3		10		5

9		6		15		4

“Magic constant should be 34. Let’s verify.” Walk through together:

- Row 1: $7 + 12 + 1 + 14 = 34$ ✓
 - Row 2: $2 + 13 + 8 + 11 = 34$ ✓
 - Rows 3, 4 — students verify independently.
 - All four columns — students verify in pairs.
 - Both main diagonals: $7+13+10+4 = 34$ ✓; $14+8+3+9 = 34$ ✓.
2. **The “extra magic” — panel sums.** “Try any 2×2 block. Does it also sum to 34?”

Pick a few:

- Top-left 2×2: $7 + 12 + 2 + 13 = 34$ ✓
- Top-right 2×2: $1 + 14 + 8 + 11 = 34$ ✓
- Centre 2×2: $13 + 8 + 3 + 10 = 34$ ✓
- Any 2×2, including those that “wrap” around the edge: also 34.

“This is a stronger property than just being a magic square. It is called **panel-magic** (or most-perfect or pandiagonal). Not every magic square has this.”

3. **A historical note.** The Khajuraho square is a real artefact — carved into a temple wall during the Chandela dynasty (~10th–11th c. CE). It is one of the earliest dated panel-magic 4×4 squares we have evidence of, anywhere. It predates Nārāyaṇa Paṇḍita (1356 CE) by several centuries; Nārāyaṇa later gave systematic methods for constructing such squares.

Honest framing: the exact dating of the Khajuraho inscription is debated among historians, but most place it in the 10th–11th century CE. The Chandela dynasty built the temple complex roughly between 950 and 1050 CE.

4. **The 4×4 construction (preview, not drilled).** The 4×4 cannot be built with the *turagagati* method. Nārāyaṇa’s *Bhadraṇita* gives a different recipe for orders divisible by 4 (called the *complement method*):

- Fill the 4×4 with 1–16 in natural order.
- Mark the cells on both main diagonals.
- For each *unmarked* cell, replace its value v with $17 - v$ (its complement to 17).

Demonstrate this on the board:

Natural-order fill:

1		2		3		4

5		6		7		8

9		10		11		12

13		14		15		16

After complementing unmarked cells:

1		15		14		4

12		6		7		9

8		10		11		5

13		3		2		16

Verify one row, one column, one diagonal — all 34. (*Note: this is a standard magic square, but not the same as the Khajuraho one, and not all such are panel-magic.*)

Activity (15 min) — Discussion + verification

- Pairs verify all eight 2×2 panels of the Khajuraho square — including the four “corner panels” formed by wrapping (top row joining bottom row, etc.).
- Class discussion: “*Why is the Khajuraho square called ‘most-perfect’?*” — student answers. Note their wording. The teacher’s frame: more constraints satisfied = more impressive.

Wrap-up (5 min)

- “Two takeaways. One: the 4×4 is built differently from the 5×5. Two: not all magic squares are equal — Khajuraho’s is unusually strong.”
- Preview Day 7: “Tomorrow we look at the mathematician who wrote down the methods: Nārāyaṇa Paṇḍita.”

Student-facing content

The Khajuraho 4×4 magic square — inscribed on a temple wall in Madhya Pradesh, ~10th–11th c. CE.

7		12		1		14

2		13		8		11

16		3		10		5

9		6		15		4

Magic constant: 34.

Extra property: every 2×2 panel also sums to 34. So do the “broken diagonals” (diagonals that wrap around the edges). This is called the **panel-magic** or **most-perfect** property — a much stronger constraint than the basic magic-square definition.

Nārāyaṇa Paṇḍita, in *Bhadragaṇita* (Book 14 of *Gaṇita-kaumudī*, 1356 CE), gave systematic methods for constructing such squares centuries later — though the Khajuraho artefact itself predates his treatise by 300+ years.

Homework

- Pick any 2×2 panel of the Khajuraho square. Verify the sum is 34. Now find a 2×2 panel that *wraps around* the edge (uses cells from both the top row and the bottom row). Verify that too. Write down both.

Differentiation

- **One tier up:** Find a 4×4 magic square that is *not* panel-magic — i.e. its rows/columns/diagonals work but some 2×2 panel does not equal 34. (Hint: the complement-method result given above.)
- **One tier down:** Just verify the basic property — rows, columns, diagonals. The panel property can come later.

Teacher notes

- Show the Khajuraho image if at all possible. Even a low-resolution photo. The artefact makes the lesson land.
- Don’t oversell the dating. “10th–11th century” is the conservative estimate; some sources claim earlier. Use the conservative range.
- The complement method shown is one of several 4×4 construction methods. Don’t drill it — Middle band only needs to know that *a* method exists.
- A few students will ask “is Sudoku a magic square?” Hold that question for Day 9. Short answer: no — Sudoku constrains rows/columns/boxes but not diagonals or sums.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — for the 4×4 construction methods.

- Material evidence: Khajuraho temple inscription (Madhya Pradesh, India), Chandela period (~10th–11th c. CE).

Day 7 — Nārāyaṇa Paṇḍita and *Gaṇita-kaumudī*

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) place Nārāyaṇa Paṇḍita (14th c. CE) in the chronology of Indian mathematics, (b) describe what the *Gaṇita-kaumudī* covers, and (c) write a short reflection on why systematic written method matters.

Materials

- Whiteboard
- Notebook
- A printed simple chronology chart (handout, see Student-facing content)

Vocabulary review

- *Gaṇita-kaumudī* (introduced in `_shared/glossary.md`) — “Moonlight of Mathematics,” Nārāyaṇa’s 1356 CE treatise.
- *Bhadragaṇita* — Book 14 of the *Gaṇita-kaumudī*, devoted to magic squares.

Lesson flow

Warm-up (5 min)

- “You’ve been using Nārāyaṇa Paṇḍita’s algorithm for three days. Today: who was he, what did he write, and why does it matter?”
- Quick free-write (60 seconds): “What do I think I know about Nārāyaṇa already?” — students jot. Not collected.

Core (20 min)

1. **Chronology.** Build a simple timeline on the board:

Period	Indian mathematics
~500 BCE	<i>Śulba Sūtras</i> — geometric ritual texts
~500 CE	Āryabhaṭa (b. 476) — <i>Āryabhaṭīya</i>
~600 CE	Brahmagupta — <i>Brāhmasphuṭasiddhānta</i>
~1100 CE	Bhāskara II — <i>Līlāvati</i> , <i>Bījagaṇita</i>
1356 CE	Nārāyaṇa Paṇḍita — <i>Gaṇita-kaumudī</i>
~1400 CE	Mādhava (Kerala) — early infinite

Period	Indian mathematics series
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“Nārāyaṇa wrote in the 14th century — relatively ‘late’ in the medieval Indian mathematical tradition, but he produced one of the most comprehensive single works.”

2. **What is in the *Gaṇita-kaumudī*?** Fourteen books (*vyavahāras*):

- Book 1–3: arithmetic (basic operations, fractions)
- Book 4–6: ratios, percentage problems, alligation
- Book 7–8: progressions and series
- Book 9: combinatorics (permutations, partitions — Nārāyaṇa was a pioneer here)
- Book 10–11: geometry of plane figures and solids
- Book 12: triangular and figurate numbers
- Book 13: practical problems (e.g. dilution of milk, mixture problems)
- **Book 14: *Bhadragaṇita* — magic squares**

“The treatise is comprehensive — it isn’t only about magic squares. But the magic-square book is the longest and most systematic in any pre-modern work we have.”

3. **Why systematic written method matters.** Discuss together:

- Before Nārāyaṇa, magic squares appeared in many Indian and non-Indian sources — but mostly as *examples*, not as recipes.
- Nārāyaṇa gives **methods**: rules that work for any input size. This is what makes it mathematics, not just a curiosity collection.
- This is the same distinction we draw today between “here’s a Sudoku puzzle” and “here’s an algorithm to generate Sudoku puzzles.”

4. **A paraphrase from the source.** From *Bhadragaṇita*, paraphrased:

“Magic squares of odd order are made by placing the first number in the top middle cell, and proceeding diagonally upward and rightward; if blocked, descend by one.”
(Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī* Book 14, 1356 CE — paraphrased.)

This is the rule you have been using. The original is in Sanskrit; this is a faithful paraphrase, not a quoted verse.

Activity (15 min) — Reflection paragraph

- Each student writes 5–7 sentences in response to:
“Nārāyaṇa Paṇḍita gave us a rule, not a magic square. Why is having a rule more valuable than having just one example? Use specific evidence from this module.”
- Pairs swap and read each other’s paragraphs. One quick comment each: “the part that landed” and “what I’d want more of.”

Wrap-up (5 min)

- Two volunteer-read paragraphs (with permission).

- Preview Day 8: “*Tomorrow we go cross-cultural — China’s Lo Shu and Europe’s Dürer.*”

Student-facing content

Nārāyaṇa Paṇḍita (14th c. CE) was a mathematician of medieval India who wrote the *Gaṇita-kaumudī* (“Moonlight of Mathematics”) in **1356 CE**. The work has 14 books (*vyavahāras*) covering arithmetic, algebra, geometry, combinatorics, and recreational mathematics.

The 14th and final book is *Bhadragaṇita* (“Mathematics of Auspicious Squares”). It is the most comprehensive pre-modern treatment of magic squares in any tradition, giving:

- The magic-constant formula $S = n(n^2 + 1)/2$ (which you derived on Day 2).
- The stair-step (*turagagati*) method for any odd-order square (which you used on Days 4–5).
- Complement-based methods for even-order squares.
- Methods for *singly-even* orders (6, 10, 14, ...).
- Discussion of equivalence and symmetry classes.

Paraphrase from the source: “Magic squares of odd order are made by placing the first number in the top middle cell, and proceeding diagonally upward and rightward; if blocked, descend by one.” (Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī* Book 14, 1356 CE — paraphrased.)

Having a *rule* is more powerful than having an *example*. A rule generates infinitely many squares. An example gives you one.

Homework

- Polish your reflection paragraph from class. Add one more specific example (e.g. “I used the rule today to construct a 7×7 square in under 5 minutes; without the rule I would have been guessing all night.”).

Differentiation

- **One tier up:** Compare Nārāyaṇa to Bhāskara II (~1150 CE). What did Bhāskara cover that Nārāyaṇa didn’t? Vice versa? (Bhāskara: calculus precursors, planetary motion. Nārāyaṇa: combinatorics, magic squares.)
- **One tier down:** Write just 3–4 sentences. Focus on the “rule vs. example” distinction.

Teacher notes

- Avoid hero-worship framing. Nārāyaṇa was a brilliant systematiser; he was not the sole discoverer of magic squares.
- A few students will ask whether Nārāyaṇa’s methods were known outside India. Answer: yes, eventually — the same odd-order method shows up in 17th-c. European mathematics (via de la Loubère, 1687) probably with intermediate Islamic-world transmission. The independent re-discovery story is interesting but speculative.

- The reflection paragraph is a low-stakes formative — don't grade harshly. Read for understanding, not for polish.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — primary source paraphrased.
- Plofker (2009), *Mathematics in India*, ch. on 14th-c. mathematics — secondary scholarly context.

Day 8 — Cross-cultural: Lo Shu and Dürer

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) identify the Chinese *Lo Shu* and Albrecht Dürer's *Melencolia I* magic squares, (b) note structural similarities and differences across cultures, and (c) begin their project planning.

Materials

- Image of the Lo Shu square (turtle-shell pattern is helpful)
- Image of Dürer's *Melencolia I* (1514) with the 4×4 corner visible
- Project planning sheet (one per pair)

Vocabulary review

- *Lo Shu* — a Chinese 3×3 magic square attested at least from the *Da Dai Liji* (Han dynasty, ~5th–1st c. BCE) and probably earlier.
- *Melencolia I* — a 1514 engraving by the German artist Albrecht Dürer; its upper-right corner contains a famous 4×4 magic square.

Lesson flow

Warm-up (5 min)

- “*Magic squares are not a one-culture phenomenon. Today we look at two famous non-Indian examples and compare.*”
- Show both images side by side, briefly. Don't analyse yet.

Core (20 min)

1. The Lo Shu (China).

The 3×3 magic square in Chinese tradition is identical (up to symmetry) to the 3×3 we've been studying:

4 | 9 | 2

3		5		7

8		1		6

Magic constant: 15 (same as our Day-1 square).

The legend: Emperor Yu (legendary, ~2000 BCE attribution) saw the pattern on the shell of a turtle emerging from the Lo river. The earliest *documented* appearance in a Chinese text is the *Da Dai Liji* (Han dynasty, roughly 5th–1st c. BCE). Treat the legend as a story; the documented record is the historical evidence.

“This is the same square — up to symmetry — as the one we built on Day 1. Two cultures arrived at the same arrangement independently, separated by huge distances.”

2. Dürer’s Melencolia I (Germany, 1514 CE).

The 4×4 in Dürer’s engraving:

16		3		2		13

5		10		11		8

9		6		7		12

4		15		14		1

Magic constant: 34.

Notable features:

- The bottom-middle two cells contain 15 and 14 — encoding the date **1514**.
- The four corners (16, 13, 4, 1) sum to 34.
- Each 2×2 panel sums to 34 (panel-magic property — same as Khajuraho!).
- Dürer’s square is not the same as Khajuraho’s, but shares the panel-magic property.

3. Compare and contrast. Build a class table:

Square	Approx. date	Culture	Order	Magic constant	Special features
Lo Shu	~5th–1st c. BCE (documented)	Chinese	3×3	15	Identical to our 3×3
Khajuraho	~10th–11th c. CE	Indian	4×4	34	Panel-magic
Nārāyaṇa’s worked examples	1356 CE	Indian	up to large n	varies	Systematic methods

Square	Approx. date	Culture	Order	Magic constant	Special features
Dürer	1514 CE	German	4×4	34	Panel-magic; encodes year

4. **Honest framing.** Magic squares appeared independently (or via transmission) in:

- **China:** Lo Shu and later expansions.
- **India:** numerous artefacts and Nārāyaṇa’s systematic methods.
- **Islamic world:** treatises by al-Bīrūnī (~11th c. CE) and others.
- **Europe:** through Islamic and (likely) Indian transmission by 14th–15th c.

We do not have a clean story of “X invented it; Y borrowed it.” Frame the topic as a *cross-cultural mathematical fascination*.

Activity (15 min) — Project planning

- Use `activities/activity-02-cross-cultural-compare.md` (a side-by-side worksheet).
- Pairs commit to their project format (poster / video / written exposition) and one or two key ideas.
- Each pair writes a 3-sentence project plan, hands in for teacher review by end of class.

Wrap-up (5 min)

- Two-pair quick share: “Our project is going to be ____ because ____.”
- Preview Day 9: project work + a brief look at Sudoku.

Student-facing content

Magic squares across cultures.

Lo Shu (China, ~5th c. BCE)

```
4 | 9 | 2
-----
3 | 5 | 7
-----
8 | 1 | 6
```

Dürer (Germany, 1514 CE)

```
16 | 3 | 2 | 13
-----
5 | 10 | 11 | 8
-----
9 | 6 | 7 | 12
-----
4 | 15 | 14 | 1
```

The Lo Shu 3×3 is the same square as our Day-1 example (up to symmetry). The Dürer 4×4 shares the **panel-magic** property with the Khajuraho square (~10th–11th c. CE) — every 2×2 panel sums to 34.

Multiple cultures arrived at magic squares: China (~5th c. BCE earliest documented), India (Khajuraho ~10th c. CE, Nārāyaṇa’s *Bhadragaṇita* 1356 CE), the Islamic world (~11th c. CE on), and Europe (~14th c. CE on). No clean single origin.

Homework

- Choose one of the three squares we've seen this fortnight (Day-1 / Khajuraho / Dürer). Verify its magic constant on a fresh piece of paper from scratch. Note one feature you find interesting.

Differentiation

- **One tier up:** Look up one more cultural example (e.g. al-Bīrūnī's magic squares, or the Jain *Pārśvanātha* yantra). Bring an image and a verified magic constant.
- **One tier down:** Focus on one comparison: how is the Lo Shu the same as our Day-1 square? Trace the rotation that takes one into the other.

Teacher notes

- Don't fall into "ancient Indians invented this first" framing. The Lo Shu is older than any dated Indian magic square we have. Nārāyaṇa's contribution is the *systematic method*, not the *first discovery*.
- The Dürer corner being a date (1514) often delights students. Use that delight to motivate the project.
- Some students will see the Khajuraho ~10th-c. dating and conclude it predates Lo Shu. That is incorrect. The Lo Shu has earlier documented evidence (Han dynasty texts). Be clear about which dates are *attestations* and which are *legends*.

Citation(s) used in this lesson

- Lo Shu: *Da Dai Liji* (Chinese, Han dynasty, ~5th–1st c. BCE) — earliest documented Chinese mention.
- Dürer, *Melencolia I* (1514) — material artefact.
- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14, 1356 CE — for context on Indian systematic treatment.

Day 9 — Sudoku Connection + Project Session

Module: Magic Squares (*Bhadragāṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) distinguish a Sudoku puzzle from a magic square (Sudoku constrains *placement* but not *sums*), and (b) make substantial progress on their module project.

Materials

- Sample 9×9 Sudoku puzzle (one per pair, easy difficulty)
- Project materials (graph paper, markers, etc.)

Vocabulary introduced today

- *Sudoku* — a 9×9 number-placement puzzle (Japanese name, from Japanese 数独). Standard rules: each row, each column, and each 3×3 block contains the digits 1–9 exactly once.
- *constraint satisfaction* — a problem type where you must place objects in a structure so that all “constraints” (rules) are satisfied. Magic squares and Sudoku are both constraint-satisfaction problems.

Lesson flow

Warm-up (5 min)

- “*Many of you have seen Sudoku. Is it a magic square?*” — take 3 student responses.
- Hold the answer. Promise: “we’ll settle this in 10 minutes.”

Core (15 min)

1. **The rules side-by-side.** Write on the board:

Property	Magic Square (3×3 , 1–9)	Sudoku (3×3 within 9×9)
Uses digits 1–9?	yes — once each	yes — once each within each 3×3 block
Each row uses each digit once?	no (a row has just 3 distinct values from 1–9)	yes (each row uses 1–9)
Each column uses each digit once?	no	yes
Diagonals sum to a constant?	yes	not required
Rows sum to a constant?	yes (= 15)	not enforced
Columns sum to a constant?	yes	not enforced

“Magic squares constrain sums . Sudoku constrains placement . Different constraints — different puzzles.”

2. **Both are constraint-satisfaction problems.** Each gives you a structure (9 cells in 3×3 , or 81 cells in 9×9) and a set of rules. You search for placements that satisfy all rules. Computers solve these the same way.
3. **A bridge example.** It is possible to have a 9×9 grid that is *both* a Sudoku AND a magic square (every row, column, *and* 3×3 block sums to 45 — which works because rows/columns automatically sum to $1+2+\dots+9 = 45$). Such squares exist; they are rare. Show one if you have time. (*Not required at Middle band.*)

Activity (20 min) — Project work

- Pairs work on their project. Teacher circulates.
- Required checkpoints by end of class:
 - The magic square is fully constructed (5×5 or larger).
 - Magic constant verified for at least 3 rows, 3 columns, both diagonals.
 - Citation is in place (Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14, 1356 CE).
 - At least one paragraph (or one slide of a video) of explanation drafted.
- The teacher does a 1-minute review with each pair to flag any errors before Day 10.

Wrap-up (5 min)

- Quick poll: “On a scale of 1–5, how ready do you feel for Day 10’s share?” Show of fingers.
- For 1–2 ratings: teacher schedules a 5-min meeting with that pair before Day 10.

Student-facing content

Sudoku ≠ magic square.

	Magic square (3×3, 1–9)	Sudoku 3×3 block
Cells	9 distinct values from 1–9	9 distinct values from 1–9
Rows sum to a constant?	Yes (15)	No requirement
Each row has all 9 digits?	No (only 3 cells)	Yes (in 9×9 Sudoku)
Diagonals constrained?	Sums	Not required

Both are *constraint-satisfaction* problems. Different constraints — different puzzles.
Sudoku appeared in modern form in the 1970s–80s (popularised in Japan from 1986).
Magic squares are at least 2500 years old (Lo Shu).

Homework

- Finish anything not finished in class. Tomorrow is the share, not the start.
- Time yourself: how fast can you construct a 5×5 magic square now, versus on Day 5?

Differentiation

- **One tier up:** Find or construct a 9×9 grid that is both a Sudoku AND a magic square (every row/column/3×3 block sums to 45). Hint: rows and columns of a Sudoku already sum to 45. You only need to also make every 3×3 block sum to 45.
- **One tier down:** Use the class time to construct one more 5×5 magic square as practice for the summative. Speed is not the goal; correctness is.

Teacher notes

- The Sudoku discussion is short on purpose. The point is to draw a clean distinction, not to teach Sudoku.
- Spend the project-work time circulating. Most errors are caught at this stage.

- Some students will worry about the “design-your-own” requirement. Remind them: choosing the size, the starting number, and the presentation format are all design choices. They are not expected to invent a new algorithm.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — paraphrased in projects.

Day 10 — Summative + Share

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) complete the summative quiz, and (b) present their original magic-square project to the class.

Materials

- Printed summative quiz (one per student) — `quizzes/summative.md`
- Project artefacts (posters, devices for videos, written exposition)
- Stopwatch (for 2-minute presentation timer)

Lesson flow

Summative quiz (15 min)

- Closed notebook. Students complete `quizzes/summative.md` individually.
- 15 min total: 3 min reading, 12 min solving.

Presentations (25 min)

- Each pair gets **2 minutes** to present + 1 minute Q&A.
- With ~10–12 pairs in a typical class, this fits in 25 min.
- Order: teacher draws pair names randomly.

What each presentation must show:

- The constructed magic square (size, magic constant verified).
- Citation of the source (Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, 1356 CE).
- One sentence on *why this particular design* — what choice did the pair make?

Wrap-up (5 min)

- One question per student on a slip of paper: “*What is one thing you’ll remember about magic squares one year from now?*”
- Collect. Read at random — 2–3 aloud.
- Close with: “*You all now know an algorithm older than most countries’ modern names. Use it. Show someone.*”

Student-facing content

Today is the share.

Bring: - Your project artefact (poster / video / written exposition). - Your two-minute presentation prepared. - Your full attention for your classmates' presentations — you will give feedback.

Reminder: **Verify before you present.** Sum at least 3 rows, 3 columns, and both diagonals out loud (or in your head). If they don't match the magic constant, fix the error *before* you stand up.

Homework — none

The module ends here. The project artefact stays with the school until the end of term as part of the module's portfolio. Reflection slips are collected.

Differentiation

- **One tier up:** Pairs who finish early help check others' projects against the rubric (specifically, the “method demonstrated correctly” criterion).
- **One tier down:** A pair that didn't quite finish presents a smaller artefact (e.g. 3×3 instead of 5×5). They still meet the citation and verification requirements.

Teacher notes

- Keep the presentations tight. 2 minutes is short on purpose — encourages preparation.
- The Q&A minute is for *one* clarifying question, not a defence. Cap it strictly.
- Score against `assessments/rubric.md` during the share — don't try to score after. Live scoring keeps things fast.
- For students who freeze: have a partner take over. The presentation is paired; both partners speak counts towards the rubric.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — referenced in project artefacts.

Module 14 (Middle) — Formative Quizzes

Two parts: Part A on Day 2, Part B on Day 5.

Part A — Day 2 (10 min, 6 questions)

Name: _____ Class: _____

1. (1 mark) What is the magic constant of a 3×3 magic square with the numbers 1 to 9?

2. (2 marks) State the magic constant formula for an $n \times n$ magic square containing the numbers 1 to n^2 . _____

3. (1 mark each) Compute the magic constant for:

(a) $n = 4 \rightarrow$ _____

(b) $n = 5 \rightarrow$ _____

(c) $n = 7 \rightarrow$ _____

4. (3 marks) Verify the magic constant by summing the rows of this 3×3 :

2 | 7 | 6

9 | 5 | 1

4 | 3 | 8

Row 1: ____ Row 2: ____ Row 3: ____

Is this a magic square? (Y / N) ____

5. (1 mark) A magic square is “magic” because:

(a) it has special religious power

(b) every row, column, and diagonal sums to the same total

(c) it uses the numbers 1 to 9

(d) it was discovered by Nārāyaṇa Paṇḍita

Circle one: ____

6. (2 marks, short answer) In your own words, explain why the magic constant for a 3×3 with 1–9 is 15. Use the number 45 somewhere in your answer.

Part A answer key

Q	Answer	Marks
1	15	1
2	$S = n(n^2 + 1)/2$	2
3a	34	1
3b	65	1
3c	175	1
4	15, 15, 15; Yes	3 (1 per row + 0 for the Y/N if rows are wrong)
5	(b)	1
6	$1+2+\dots+9 = 45$; split among 3 rows $\rightarrow$	2

Q	Answer	Marks
	15. (Wording variants accepted if both numbers appear and the logic is right.)	

Total Part A: 12 marks.

Part B — Day 5 (15 min, 5 mixed problems)

Name: _____ **Class:** _____

1. (2 marks) State the *turagagati* (stair-step) rule in your own words. Five steps, one line each.

Step 1: _____ Step 2: _____

_____ Step 3: _____

_____ Step 4: _____

_____ Step 5: _____

2. (4 marks) Construct a 3×3 magic square from scratch using the *turagagati* method. Show the order in which you placed each number (label 1 through 9 in placement order on the grid). Verify one row.

—		—		—
—		—		—
—		—		—

Verification: one row sums to _____.

3. (4 marks) Construct a 5×5 magic square using the *turagagati* method. Verify the magic constant by summing the first row and the main diagonal.

(Use the grid on the back of this sheet.)

First-row sum: _____ Main-diagonal sum: _____

4. (2 marks) True or false?

(a) The magic constant of an $n \times n$ magic square is always $n(n^2 + 1)/2$ — even if the numbers in the square are not 1 to n^2 . [T / F]

(b) The 3×3 magic square with 1–9 is unique up to symmetry. [T / F]

5. (3 marks, short answer) Nārāyaṇa Paṇḍita lived in the 14th century CE. Why does the date of his treatise (1356 CE) matter? What's the difference between *discovering* a magic square and *describing the method* to build one?

Part B answer key

Q	Answer	Marks
1	5 steps: (1) 1 in top-middle; (2) up-right diagonal for next; (3) off-top → wrap to bottom; (4) off-right → wrap to left; (5) blocked → directly below.	2 (any reasonable rephrasing)
2	Standard 3×3 square; row sums = 15	4 (3 for correct square; 1 for verified row)
3	5×5 magic square; first row = 65; diagonal = 65	4 (2 for correct square; 1 each for the two sums)
4a	F (the formula assumes numbers 1 to n^2)	1
4b	T	1
5	The formula's date matters because it shows Nārāyaṇa systematised the method 300+ years before the same method appeared in European mathematics (de la Loubère, 1687). The difference between <i>discovering</i> and <i>describing the method</i> : discovery gives you one example; method gives you a recipe that generates infinitely many.	3 (1 for date awareness; 1 for discovery vs method distinction; 1 for clarity)

Total Part B: 15 marks.

What to do with the results

- Class average below 8/15 on Part B = repeat the stair-step demonstration in next warm-up; require additional practice in homework.
- Individual below 5/15 = pull aside for 10-minute reteaching of step 5 (the most common error point).

Module 14 (Middle) — Summative Quiz

Time: 15 min · **Marks:** 25 · **Closed notebook**

Instructions: Name + class at top. Show your work on construction problems. No calculators.

Name: _____ **Class:** _____

Section A — Definitions ($4 \times 1 = 4$ marks, 2 min)

1. What is a magic square? _____
 2. What is the magic constant? _____
 3. Write the magic constant formula: _____
 4. Name the Sanskrit term for the stair-step method (Nārāyaṇa's): _____
-

Section B — Apply the formula ($5 \times 1 = 5$ marks)

Compute the magic constant for each:

- (a) $n = 3 \rightarrow$ _____
 - (b) $n = 4 \rightarrow$ _____
 - (c) $n = 5 \rightarrow$ _____
 - (d) $n = 6 \rightarrow$ _____
 - (e) $n = 8 \rightarrow$ _____
-

Section C — Verify (4 marks)

5. Here is a candidate 3×3 grid:

2		9		4

7		5		3

6		1		8

- (a) Sum each row: _____, _____, _____ (1 mark)
 - (b) Sum each column: _____, _____, _____ (1 mark)
 - (c) Sum both diagonals: _____, _____ (1 mark)
 - (d) Is this a magic square? (Y / N) _____ (1 mark)
-

Section D — Construct (5 marks)

6. Construct a 5×5 magic square using the *turagagati* method on the grid below. Verify by computing the first-row sum.

—		—		—		—		—
—		—		—		—		—
—		—		—		—		—
—		—		—		—		—
—		—		—		—		—

First-row sum: ____ (should be 65)

Section E — Historical context (3 marks)

7. (1 mark) Who wrote *Bhadraganita*, the most extensive pre-modern treatment of magic squares? _____

8. (1 mark) In what year (approximately) was that treatise written? _____

9. (1 mark) Name one other culture in which magic squares appeared. _____

Section F — Open response (4 marks)

10. A friend says: “*Magic squares are just a coincidence — pretty patterns with no real math.*” In 4–6 sentences, respond to your friend. Use the words **algorithm**, **constraint**, and **rule** somewhere in your answer.

Answer key

Q	Answer	Marks
1	An $n \times n$ grid of distinct numbers where every row, column, and main diagonal sums to the same total.	1
2	The common sum (rows = columns = diagonals = this total).	1
3	$S = n(n^2 + 1)/2$	1
4	<i>turagagati</i>	1
5a	15	1
5b	34	1
5c	65	1
5d	111	1

Q	Answer	Marks
5e	260	1
6a	15, 15, 15	1
6b	15, 15, 15	1
6c	15, 15	1
6d	Y	1
7 (square)	Correct 5×5 (the standard <i>turagagati</i> output)	4 (deduct 1 per row error, capped at 4)
7 (verify)	first-row sum = 65	1
8	Nārāyaṇa Paṇḍita	1
9	1356 CE (accept “14th century”)	1
10	China (Lo Shu), Islamic world, Europe (Dürer) — any one	1
11 (open)	A complete answer uses <i>algorithm</i> (we have a method, not luck), <i>constraint</i> (rows/columns/diagonals must all match), <i>rule</i> (the stair-step gives an exact rule); makes one specific point grounded in the module.	4: 2 for using all 3 words; 2 for substantive argument.

Total: 25 marks.

Grading bands (combined with project for module score out of 50): - 40–50: Exceeds - 30–39: Meets - 20–29: Approaches - 0–19: Beginning

Homework

Homework — Module 14 Magic Squares (Middle)

Day 1 → Day 2

Task: Copy the 3×3 magic square from class. Beside it, draw three rotations (90°, 180°, 270°). Verify one row of each.

Time: 10 min.

Day 2 → Day 3

Task: Verify the magic constant formula $S = n(n^2 + 1)/2$ for $n = 8$ and $n = 9$. Show your arithmetic. (Answers: 260 and 369.)

Time: 8 min.

Day 3 → Day 4

Task: Take the 3×3 magic square and add 10 to every entry. The square becomes one with numbers 11–19 instead of 1–9. What is the new magic constant? Verify one row.

Time: 10 min.

Day 4 → Day 5

Task: Construct three 5×5 magic squares from scratch using the *turagagati* method. Verify the magic constant of each (should be 65). Yes, they should all be the same square — that's part of the discovery.

Time: 20 min.

Day 5 → Day 6

Task: Construct a 7×7 magic square using the stair-step method. Magic constant should be 175. Bring it tomorrow for spot-checks.

Time: 25 min.

Day 6 → Day 7

Task: Pick any 2×2 panel of the Khajuraho square (including one that wraps the edge). Verify the sum is 34. Write down both the panel cells and the sum.

Time: 5 min.

Day 7 → Day 8

Task: Polish your in-class reflection paragraph on Nārāyaṇa. Add one more specific example from your own construction work.

Time: 15 min.

Day 8 → Day 9

Task: Submit your project plan (4–6 sentences): format, target audience, the magic square you'll construct, what is unique about your design.

Time: 15 min.

Day 9 → Day 10

Task: Finalise project artefact and rehearse the 2-minute presentation aloud at least once. Time yourself.

Time: 30 min (cap).

Writing Prompts

Writing Prompts — Module 14 Magic Squares (Middle)

Six prompts. Useful as exit tickets, homework reflections, or project warm-ups.

Reflection prompts

- 1. (Day 1, end of class, 3 min)** When you tried to build the 3×3 magic square by trial and error, what was going through your head? Did you have a strategy, or were you just guessing? Be specific.
- 2. (Day 7, end of class, 5 min)** You spent three days using Nārāyaṇa's *turagagati* algorithm before learning who Nārāyaṇa was. Did learning about him change how you feel about the method? (Either answer — yes, no, somewhere in between — is fine, but explain it.)

Explanatory prompts

- 3. (Day 4 or homework, ~120 words)** Explain the *turagagati* method to a Grade 5 student who has never seen it. Use no more than 7 sentences. Include one specific example.
- 4. (Day 6 or homework, ~120 words)** Explain *why* the Khajuraho 4×4 square is called “panel-magic.” Use the word **constraint** in your answer.

Argumentative prompt

- 5. (Day 8 or homework, 1 paragraph)** Some sources call magic squares “Indian magic” or “Chinese magic” depending on the culture. Others insist that magic squares appeared independently in many cultures, and no single culture “owns” them. Pick a position and defend it in 5–7 sentences. Use what you know about Lo Shu, Khajuraho, Nārāyaṇa, and Dürer.

Creative prompt

- 6. (Day 7 or Day 8, optional, 1 page)** Write a short dialogue (50–150 words) between Nārāyaṇa Paṇḍita (1356 CE) and Albrecht Dürer (1514 CE), meeting across time. They show each other their magic squares. Let them disagree on something — maybe about *why* they care about these patterns. Don't have them magically agree.

Prompt-to-band notes

- For **Primary** band, prompts 1, 3, and 6 work as drawings or oral responses.
- For **Senior** band, prompt 5 deserves a full essay treatment with sources cited from `sources.md`.

How to give feedback

- Prompts 1, 2: acknowledge with one sentence — no correction.

- Prompts 3, 4: respond with “the part that landed” and “the part that confused me.”
- Prompt 5: use the rubric — does the writing acknowledge the other side, give a specific example, reach a clear position?
- Prompt 6: no grading. Read the best ones aloud (with permission).

Activity 1 — Class 3×3 Magic-Square Tournament

Module: Magic Squares — Bhadragnaṇita · **Band:** Middle · **Duration:** 25 min

Purpose

Reinforce the day-by-day concepts of Magic Squares — Bhadragnaṇita through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

grids on paper, pencils, timer

What students do

In pairs, students race to construct the 3×3 magic square with 1–9 from a starting 9-position permutation. First pair with correct sums wins.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (15 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Activity 01 — Stair-Step Drill

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Day:** 4 · **Time:** 15 min

Type

Individual + paired practice

Goal

Each student constructs three 3×3 magic squares using Nārāyaṇa's *turagagati* method, with the partner checking accuracy.

Materials per student

- 3 printed blank 3×3 grids
- The 5-step rule sheet (from Day 4 student-facing content)
- Pencil

Set-up (teacher, before class)

- Print blank 3×3 grids (3 per student).
- Have the 5-step rule visible on the board.

Instructions (student-facing handout)

Stair-Step Drill — work individually first, then in pairs.

Round 1 (5 min) — Slow build, rule sheet open. Construct a 3×3 magic square using the *turagagati* method. Take your time. Speak the steps out loud quietly. Verify all three row sums equal 15 before stopping.

Round 2 (4 min) — Faster, rule sheet open. Construct another 3×3 magic square from scratch. Time yourself.

Round 3 (3 min) — Rule sheet closed. Construct one more 3×3 , this time from memory. Time yourself.

Pair check (3 min) — swap sheets with your partner. Did they get the same square? (They should — there is only one 3×3 magic square up to symmetry, and the algorithm always produces the same one.) Compare round-by-round times.

What to expect

All three squares should be identical:

2		7		6

9		5		1

4		3		8

If a student's square is different, walk through their steps to find where the rule was misapplied — usually step 5 (“blocked → directly below”).

Discussion prompts (last 2 min)

- Did you get the same square all three times? If yes, why? If no, where did the difference come from?
- What did Round 3 (from memory) feel like compared to Round 1?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Student narrates “1 in top middle, then up-right, off the top so wrap to bottom...”	“Where does 6 go? Up-right from 5 — that’s outside the top, so wrap to...”
Needs intervention	Student stares at the blank grid; doesn’t know where to begin	Silence; “I don’t get it.”

Safety / sensitivity

- No timed competition between students. Each student times *themselves*.

Differentiation

- **Tier up:** Construct a 7×7 magic square using the same method. Constant should be 175. Verify two rows and the main diagonal.
- **Tier down:** Stay on the 3×3 . The rule must be automatic before scaling up.

Clean-up

- All 3 constructed squares stay in the student workbook for later reference.

Activity 02 — Cross-cultural Comparison

Module: Magic Squares (*Bhadraganita*) · **Band:** Middle · **Day:** 8 · **Time:** 15 min

Type

Paired worksheet with discussion

Goal

Pairs compare three magic squares from three different cultures and identify the structural features they share or differ on.

Materials per pair

- 1 comparison worksheet (template below)
- Printed images of the three squares (or projected on the board): Lo Shu, Khajuraho, Dürer

Instructions (student-facing)

Cross-Cultural Comparison.

You will compare three magic squares: 1. **Lo Shu** (China, ~5th–1st c. BCE attested) 2. **Khajuraho** (India, ~10th–11th c. CE) 3. **Dürer** (Germany, 1514 CE)

Complete the table together. For each cell that asks about a sum, verify the sum yourself (don't take the question's word for it).

Comparison worksheet

Question	Lo Shu (3×3)	Khajuraho (4×4)	Dürer (4×4)
Order n	3	4	4
Magic constant	_____	_____	_____
Numbers used	1–_____	1–_____	1–_____
First row	, , _	, , ,	, , ,
Centre number(s)	_ (one cell)	, , , (four cells form the centre 2×2)	, , ,
Sum of corners	_ + _ + _ + _ = _	_ + _ + _ + _ = _	_ + _ + _ + _ = _
Every 2×2 panel sums to constant? (panel-magic?)	(irrelevant — no 2×2 panels in 3×3)	Yes / No	Yes / No
Approximate date	~5th–1st c. BCE	~10th–11th c. CE	1514 CE

Discussion (last 3 min)

- Which square is **most constrained** (i.e. satisfies the most rules)? Why?
- Which is **simplest** to construct? Which would you struggle with most?
- Do you see any reason the same idea appeared in three different places?

Reference data

Lo Shu (China):

```
4 | 9 | 2
3 | 5 | 7
8 | 1 | 6
```

Magic constant: 15

Khajuraho (India):

```
7 | 12 | 1 | 14
2 | 13 | 8 | 11
16 | 3 | 10 | 5
9 | 6 | 15 | 4
```

Magic constant: 34

Dürer (Germany):

16		3		2		13
5		10		11		8
9		6		7		12
4		15		14		1

Magic constant: 34

Answer key (teacher)

- Lo Shu magic constant: 15. Numbers 1–9.
- Khajuraho: 34. Numbers 1–16. **Panel-magic: Yes** (every 2×2 sums to 34).
- Dürer: 34. Numbers 1–16. **Panel-magic: Yes** (every 2×2 sums to 34).
- Both Khajuraho and Dürer are panel-magic; this is unusual.

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Pair is summing cells in pencil, comparing answers	“Wait — let me check the top-left 2×2 : $7+12+2+13 = 34$ ✓”
Needs intervention	One partner fills in the worksheet; the other watches	Silence; “you do it”

Differentiation

- **Tier up:** Compare the four corners of each square. Do they sum to the magic constant? (Not in general, but yes for Dürer and Khajuraho — both are “associative” in this sense.)
- **Tier down:** Just verify the basic magic constant on one square. Skip the panel-magic question for now.

Project Brief

Project Brief — Module 14 Magic Squares (Middle)

Work sessions: Days 8 + 9 · **Presentation:** Day 10 · **Total marks:** 25 (project 20 + presentation 5)

The project

Each pair (or individual, with teacher approval) designs and presents an **original magic square** of order 5×5 or larger. Choose ONE of three formats:

Format 1: Labelled poster

Produce an A3-sized (or larger) poster that shows: - The constructed magic square (at least 5×5)
 - Step-by-step construction labels showing the *turagagati* placement order (number each cell 1, 2, 3, ... in the order the algorithm placed it) - A verification panel showing row, column, and

diagonal sums - A 50-word caption stating the magic constant and citing Nārāyaṇa Paṇḍita's *Gaṇita-kaumudī*

Format 2: 60-second demo video

Make a short video that walks a Grade 5 viewer through constructing a 5×5 magic square. Constraints: - Total length: 60 seconds (± 10 sec) - Must show: the placement of at least cells 1, 2, 3, and one “blocked $\rightarrow$ directly below” step - Must include verbal verification of the magic constant - Cite the source verbally in the closing line (e.g. “Method from Nārāyaṇa Paṇḍita's *Gaṇita-kaumudī*, 1356 CE”)

Format 3: Written exposition

Produce a 2-page written piece that: - Constructs an original magic square (5×5 or larger) - Explains the construction in clear prose, step-by-step - Shows the verification arithmetic - Includes a paragraph on Nārāyaṇa Paṇḍita and the *Gaṇita-kaumudī* - Cites the source in a closing line

Design choices

You are not expected to invent a new algorithm. Your design choices are: - **Order (size)**. Pick 5×5 , 7×7 , or 9×9 . - **Starting cell**. The classical method starts in the top-middle. What happens if you start elsewhere? (You may use any valid starting cell — bottom-middle, top-right, etc. — and your magic square will be a rotation/reflection of the classical one.) - **Number range**. Default is 1 to n^2 . For tier-up, try a square with consecutive numbers from a different starting number (e.g. 10 to 34 for $n = 5$). Compute the new magic constant. - **Presentation format and style**. This is where most of your originality goes.

Why this matters

You're moving from *using* an algorithm to *teaching* an algorithm. Teaching is the proof of understanding. The verification step is what makes your work mathematics rather than craft.

What success looks like

A strong project will: - Demonstrate the *turagagati* method accurately (no arithmetic errors). - Be pitched for an audience (Grade 5 for the video, peer reader for poster/exposition). - Show your own design choices. - Include the verification panel — without exception.

Working in pairs

- Pair work is the default.
- Individual work allowed for any format with teacher approval.
- Groups of 3 acceptable — each member must have a distinct role (e.g. designer, constructor, verifier).

Citation rules

- At minimum **one** citation must appear visibly in the project.
- Format: “Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE.”

- The source must be cited even if the artefact is a video — say it aloud at the end.

Self-reflection (submitted with project on Day 10)

In 4–6 sentences: 1. What did you set out to make, and what did you end up with? 2. What’s the most surprising thing you learned while making this? 3. If you had three more days, what would you do differently?

Rubric

Project Rubric — Module 14 Magic Squares (Middle)

Project marks: 20 (out of 25 with presentation 5).

Criteria

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
Method demonstrated correctly	Stair-step placement order shown clearly; algorithm applied without error; the resulting square is verifiably magic.	Method shown; one minor error in placement OR in verification .	Method partially shown; multiple errors or significant gaps.	Method incorrect, absent, or simply copied.	× 2
Magic constant verified	All rows, columns, and both diagonals verified; arithmetic correct.	At least 3 rows, 3 columns, both diagonals verified.	Some verification present; incomplete.	No verification shown.	× 1.5
Use of sources	Cites Nārāyaṇa Paṇḍita’s <i>Gaṇita-kaumudī</i> (Book 14, 1356 CE) in the correct format; the citation appears	Citation present and correct.	Citation present but vague or incomplete.	No citation, or wrong source.	× 1

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
Originality of design	prominently. The pair makes a clear, defensible design choice (size, starting cell, number range, or presentation style) and explains it.	Clearly the pair's own work; one design choice made.	Largely a class example reproduced.	Copied verbatim.	× 1
Communication / craft	Visuals or narration are clear, deliberate, polished. Arithmetic shown is correct and legible.	Communicates clearly with minor craft issues.	Message comes through with effort.	Hard to follow.	× 0.5

Maximum: 24. Project score = (sum × weights) capped at 20.

Presentation (5 marks)

Criterion	4	3	2	1	Weight
Within time (2 min)	1:50–2:10	1:30–2:30	1:15–2:45	Outside	× 0.5
Both partners contribute	Equal	Both speak unequally	One dominates	One silent	× 0.5
Clarity	Confident, audible	Mostly clear	Audible in front rows only	Mostly inaudible	× 0.25

Maximum: 5.

Format-specific notes

- **Poster projects:** the poster is the artefact; the 2-min presentation walks the class through the cleverest design choice.

- **Video projects:** the 60-second video itself is the artefact; the 2-min presentation explains the design choices (“we used this example because ...”).
- **Written exposition:** the 2-page document is the artefact; the 2-min presentation summarises and reads the most important sentence aloud.

Combined module score

Component	Marks
Summative quiz	25
Project	20
Presentation	5
Total	50

Grading bands: - 40–50: Exceeds - 30–39: Meets - 20–29: Approaches - 0–19: Beginning